

Full title: Comparison of rowing performance improvements following various high intensity interval trainings

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ABSTRACT

To examine the effects high intensity interval training (HIT) and supramaximal interval training (SMIT) on indoor rowing performance; 20 male lightweight collegiate rowers (age= 21.77±2.35 years, height= 178.4±6 cm, body mass= 69.6±3.1 kg,) participated in the study. Baseline testing involved a 2000 m time trial and incremental exercise test to determine VO_2Peak , peak power output (PPO) and power at 4 mmol.L⁻¹ blood lactate threshold. After the baseline tests, participants were allocated to SMIT or HIT intervention groups, which they performed 8 times over a 4 weeks period (2 x week, 2 days apart). The SMIT involved 10 x 30 s intervals at 150% of the PPO with 4 min rest. The HIT involved 8 x 2.5 min intervals at 90% of the PPO with 3 min rest. 5.7s and 5s of improvements were observed in 2000 m. performance after SMIT and HIT interventions respectively. 2000 m time trial performance, 2000 m power, PPO, relative and absolute VO_2Peak were significantly improved after both training interventions. However; the differences between the groups were not significant. As a result, 4 weeks of SMIT improves 2000 m rowing ergometer performance and related physiological variables in a similar fashion with HIT in collegiate rowers.

Keywords: ergometer; 2000 m time trial; Endurance; Peak Power Output; VO_2Peak ; supramaximal interval training

1 INTRODUCTION

2 Competitive rowing is a sport that requires highly developed aerobic and anaerobic
3 capacities. The energy needed for a 2000-m ergometer rowing was estimated to be 65-75%
4 aerobic and 25-35% anaerobic (10). Anaerobic power is specifically crucial during the initial
5 sprint and final phases of the competition and characteristically ensures 20-30% of the energy
6 requirement of a 2000-m race (25, 31). 2000 m rowing ergometer time trial is the most
7 common measure of rowing performance and has become an important selection tool for
8 national rowing organizations (31). By testing rowers individually through competition-length
9 time trials on ergometers, coaches are able to evaluate each athlete's ranking on the team in a
10 controlled environment. When comparing on-water to simulated rowing methodologies only
11 slight disparities in physiological responses exist (19, 20).

12 High intensity interval training (HIT) and supramaximal interval training (SMIT) are both
13 considered to be commonly used performance enhancing methods in individual and team
14 sport training programs. HIT involves work at the speed or work rate corresponds to the 90%
15 to 100% of velocity at VO_{2max} . Work-rest ratios applied during HIT sessions vary from 1:1 to
16 2:1. SMIT involves the work at intensities greater than 100% of VO_{2max} , often with work-rest
17 ratios from 1:3 to 1:9 (2, 11).

18 Greater improvements following HIT intervention compared with traditional continuous
19 training on rowing time trial performance were reported in previous studies (8, 9).
20 Additionally, several studies demonstrated some benefits on VO_{2Peak} and short term power
21 output after SMIT (11, 26). However, to our knowledge, no single study has examined the
22 effects of SMIT on rowing time trial performance. Thus, the aim of the current study is to
23 compare the effects of 4 weeks of HIT and SMIT interventions on 2000 m rowing time trial
24 performance and the parameters related to rowing performance such as peak power output
25 (PPO), VO_{2Peak} and power at 4 mmol.L⁻¹ lactate level.

1 **METHODS**

2 **Experimental Approach to the Problem**

3 To compare the effectiveness of different training regimens on indoor rowing performance,
4 lightweight collegiate rowers were tested for 2000 m time trial and maximal incremental
5 exercise performances initially. After the baseline measures, subjects were matched according
6 to their PPO value obtained from the incremental exercise test and then randomly assigned
7 into two different training regimens as HIT and SMIT for 4 weeks (2 sessions per week for 4
8 weeks, 2 days between sessions, 8 sessions total). The time between the end of pre-testing and
9 beginning of the training was 48 hours. After 4 weeks of training, all baseline measures were
10 retested at week 5 to compare the improvements in 2000 m time trial test. Post-testing was
11 employed after the completion of training interventions with 48 hours of rest before the tests.
12 Training interventions were conducted in the last mesocycle of the specific preparation period
13 of the squad's yearly training plan when all athletes performed rowing ergometer exercises
14 regularly.

15 Participants were asked to refrain from any food intake for 3 h before the measurements,
16 and to avoid caffeine and alcohol intake and strenuous exercise for 48 h before tests. All
17 participants reported that they have not consumed any nutritional or ergogenic supplements
18 excluding multivitamin pills for the previous 6 weeks period and the same routine was
19 followed during the training intervention period. Participants were also instructed to keep a
20 diary of dietary intake on the day before the baseline tests, and the same dietary intake was
21 replicated on the post-intervention test (8, 9). Tests were conducted approximately at the same
22 time of the day (± 1 h) for each participant, and rowers used the same ergometer during the
23 tests.

Participants were asked to keep a training diary during the whole study time. Training diaries consisted of type of training, duration, HR and distance or mean power (where appropriate) of each training. Intensity ratings of each training session were provided by the participants themselves using CR10 Borg Scale (4) for Rating of Perceived Exertion (RPE). To quantify the amount of total training that was completed in both HIT and SMIT interventions, the current study used a method focusing on the session RPE. The session RPE method provides a mechanism for quantifying the exercise intensity component and was reported as a reliable tool for rowers and other athletic populations (7) . In order to calculate training amounts and loads using Session RPE method, RPE of each training session was multiplied by the duration (minutes) of the session. Participants were the part of the same squad which performs the same training program, thus only difference in training between the groups was the ergometer protocol they were undertaking twice a week.

Participants

Twenty male, national level, lightweight collegiate rowers (age = 21.77 ± 2.35 years, height = 178.4 ± 6 cm, body mass = 69.6 ± 3.1 kg, sport age = 4.3 ± 2.4 years, $VO_{2Peak} = 56.6 \pm 5.7$ ml.kg.min⁻¹) voluntarily participated in the study. The study was approved by the institutional human research ethics committee. Prior to the tests, participants were provided with a sheet which contained information about the study design and any possible risks, and they signed written informed consent forms to participate in this study.

Procedures

2000 m. Time Trial Test

1 An air-braked rowing ergometer (model D, Concept II, Morrisville, VT, USA) was used
2 for rowing performance measures. Drag factor settings of the ergometers were adjusted to 135
3 as recommended by Amateur Rowing Association (ARA) for lightweight men rowers (28).
4 For the time trial test, participants were asked to perform an all-out 2000-m on ergometer.
5 Heart rate (HR) was recorded with a telemetric HR monitor throughout the test (s610i, Polar
6 Electro Oy, Finland). Completion time, stroke rate, HR, average power outputs were recorded
7 immediately after the test for the whole test and each 500 m splits separately. Test-retest
8 reliability of 2000 m rowing ergometer time trial was previously reported with a coefficient
9 of variation (CV) of 1% (17, 29).

10 *Incremental Exercise Test*

11 To determine the metabolic responses to loading, incremental rowing ergometer test
12 recommended by Australia Sport Institute (AIS) was executed (13). The test protocol was
13 discontinuous with progressive 4 min increments, consisting of six submaximal stages and a
14 final (7th) maximal stage. The stages were separated by 1 min recovery intervals during which
15 blood samples for lactate analyses were taken. The work intensity for the submaximal stages
16 was determined individually based on each participant's best time during 2000 m time trial
17 test. The average 500 m pace of the 2000 m maximal test time plus 4 sec was calculated, to
18 give the pace (per 500 m) that the participant was required to maintain during the sixth stage
19 of the test. Successive amounts of 6 sec per 500 m were added in order to calculate the
20 required pace for the earlier work intensity. The final (7th) stage performed with 4 min
21 maximal effort.

22 Gas exchange during the test was measured breath by breath with a gas analysis system
23 (Oxycon Mobile, Jaeger GmbH, Germany). HR was recorded during test via heart rate belt
24 (Polar Electro OY, Finland). Blood lactate concentrations were measured using an automated

1 lactate analyser (YSI Sport 1500, Yellow Springs, Ohio, USA). Prior to tests lactate and gas
2 analysers were calibrated according to manufacturers' instructions.

3 VO_2Peak was determined by averaging the 4 highest consecutive oxygen consumption
4 value recorded during the last stage of the test. Mean power output in the last stage of the
5 incremental test was deemed as Peak Power Output (PPO), and was used to determine the
6 work intensity at which each participant would row during the HIT and the SMIT sessions. As
7 recommended by AIS (13), the 4 mmol.L⁻¹ fixed blood lactate concentration method was used
8 to determine anaerobic threshold (AT). Values at the fixed blood lactate concentration of 4
9 mmol.L⁻¹ were determined via a software package (Lactate-E, v2.0, National University of
10 Galway, Ireland) (27).

11 12 *High Intensity Interval Training Protocol*

13 The HIT protocol used in the current study was adapted from the study of Driller et al(8).
14 The group completed 8 sessions in 4 weeks (2 sessions per week, 2 days between each
15 session). Each participant rowed at 90% of their individual PPO determined during 2000 m
16 time trial test, for 2.5 min per set, and completed 8 sets per training. Participants were
17 instructed to continue rowing at 40% of their PPO for 3 min for recovery. Intensity of the
18 recovery period was chosen so as to promote the efficient rate of lactate removal and rapid
19 HR recovery. Each session lasted approximately 45 min.

20 21 *Supramaximal Interval Training Protocol*

22 The SMIT group completed 8 sessions in 4 weeks (2 sessions per week, 2 days between
23 each session). Each training consisted of 10 sets of 30 sec efforts with the velocity of 150% of
24 the PPO determined via an incremental exercise test. Various work-rest ratios from 1:3 to 1:9

were suggested for SMIT trainings in the literature. In order to match the exercise duration with the HIT intervention work-rest ratio was selected as 1:8 and participants were instructed to continue rowing at 40% of their PPO for 4 min for recovery (2, 5, 11) Intensity of the recovery period was chosen due to promote the efficient rate of lactate removal and rapid HR recovery. Each session lasted approximately 45 min.

Statistical Analysis

Results of the measured variables (2000 m time, 2000 m power, $\text{VO}_{2\text{Peak}}$, PPO and 4 mmol.L⁻¹ power) and percent changes of the same variables were compared using paired *t* tests for each training regimen and independent *t* tests were used to analyze training diary data, using the SPSS (version 20.0; SPSS, Inc., Chicago, IL) with a statistical significance of $P < 0.05$. Standardized changes in mean of each measure were used to assess magnitudes of effects and provide the likelihood of the true effects being practically positive, trivial and negative by dividing the changes by the smallest worthwhile change. The mean effects of training and their 90% confidence limits were estimated using a Microsoft Excel Spreadsheet (3). The likelihoods were set as $< 1\%$, almost certainly not; 1-5%, very unlikely; 5-25% unlikely or probably not; 25-75%, possibly or maybe; 75-95%, likely or probably; 95-99%, very likely; $> 99\%$, almost certainly. To summarize, likelihood values between 0-25% was deemed as negative, 25-75% was deemed as trivial and $> 75\%$ was deemed as positive effect in the tables that presented in the results. Measures of reliability known as CV was used as the smallest worthwhile change of each variable (16). CV for the mean power during 2000 m time trial was reported as 2%, the CV reported for time to complete a 2000 m time trial in the same study was 1% (29). The CVs for $\text{VO}_{2\text{Peak}}$, 4 mmol.L⁻¹ power output and PPO were reported as 2.2%, 1% and 2% respectively. All of these CV values were halved for each variable and used as the smallest worthwhile change in the current study. Thus, the

worthwhile change was accepted as 1% for 2000 m power, 0.5% for 2000 m time, 1.1 % for VO₂Peak, 0.5 % for 4 mmol.L⁻¹ power and 1% for PPO, respectively (8).

RESULTS

Table 1 presents the raw data that was collected before and after the exercise interventions. The clinical significance that SMIT had on performance and the chances that SMIT had a positive, negative or trivial effect on performance were also provided in Table 1. According to the *t* test results, all values were significantly improved as a result of the 4-week-training intervention compared to the pre-test results in both groups ($P < 0.01$). However, none of the variables showed any practically significant difference between the SMIT and HIT groups. The likelihood that the SMIT was beneficial compared with the HIT group was $\geq 36\%$ and $\geq 44\%$ for the 2000 m power and PPO variables respectively. The results, determined for the 2000 m time trial result, absolute VO₂Peak and relative VO₂Peak were found $\geq 39\%$, $\geq 32\%$ and $\geq 34\%$, respectively.

TABLE 1 NEAR HERE

As displayed in Figure 1, percentage improvements in 2000 m time trial performance, 2000 m power output, peakVO₂, PPO and 4 mmol.L⁻¹ power values were similar and not significantly different between both groups (P=0.94, P= 0.93, P= 0.88, P=0.91 and P= 0.95, respectively).

FIGURE 1 NEAR HERE

According to the results of the training diary analysis there were no significant differences (p=0.96) in the total amount of the training performed between the groups during the whole study process (as identified by the session RPE method, SMIT= 8813 ± 7380; HIT= 8477 ± 7654).

DISCUSSION

To our knowledge, no previous study has compared the effects of SMIT and HIT on rowing performance. The current study is the first that demonstrated the effects of SMIT upon the performance of rowers showing that SMIT improves performance in a similar fashion with HIT in lightweight collegiate rowers.

The rowing performance differences after SMIT and HIT interventions were not statistically or practically significant. Regardless of training group, all values of the measured variables were significantly improved between pre and post test (Table 1). Besides, performance improvements as a percentage in 2000 m time trial time, PPO, 2000 m power, VO_2Peak and 4 mmol.L^{-1} power were all similar for both training interventions (Figure 1).

Significant improvements in time trial performances were demonstrated in runners and cyclists following HIT (21). However, the research data in the literature on the effects of HIT on rowing performance is limited. Performance improvements of 3% were determined in 5 km (18) and 10 km (1) running times in middle and long distance runners after HIT intervention. In a study conducted on highly trained cyclists, improvements in 40 km time trial performance were reported to be between 4.4% and 5.8% after similar HIT protocols (23). It was found that performance enhancements related to rowing time trial performance may be associated with improvements in PPO, 4 mmol.L^{-1} power and VO_2Peak values (19, 33). These improvements were demonstrated also in the current study for both training interventions.

Improvements between 5% and 15% in VO_2Peak following various HIT interventions were reported in several studies (12, 23, 32). Similarly, 7% improvement in VO_2Peak was determined in runners, swimmers and cyclists after HIT intervention (21). In a study

1 conducted on highly trained runners using a similar HIT method, significant improvements of
2 VO_2Peak around 5% were noted (32). Moreover, 8% improvement in VO_2Peak after 4 weeks
3 of HIT was noted for highly trained cyclists (23). Improvements of about 5% in VO_2Peak
4 were noted in the current study following SMIT and HIT interventions. Though genetics and
5 initial fitness level can be considered as important factors influencing improvements in
6 VO_2Peak , training stimulus related to HIT and SMIT can also be labelled as a determining
7 factor which contribute to a magnitude of the enhancement (22). The findings in various
8 research (32) about the effectiveness of training at or close to the velocity corresponding to
9 VO_2Peak are supported in the current study when the improvements after the HIT intervention
10 was taken into account. In a study that used similar SMIT intervention, the change in
11 performance of 3000 m running following SMIT was found greater compared with the one in
12 continuous running (5). In the same study, after 6 weeks of training, improvements of about
13 6.2% for males and 10.5% for females were determined in 3000 m running performance
14 following SMIT intervention and HIT intervention did not produce significantly greater
15 improvement in endurance variables than SMIT. In addition, it has been established that
16 SMIT enhances anaerobic variables significantly greater than HIT.

17 Lactate clearance rate at a given submaximal power output, and ability to elicit less lactate
18 with a higher work rate are important predictors of rowing performance due to the
19 relationship between blood lactate levels and muscle respiratory capacity(19). Work output at
20 4 mmol.L^{-1} blood lactate level was established as an important predictor of rowing
21 performance in several studies (6, 19, 30)Improvements of about 7.5% and 6.3% in power at 4
22 mmol.L^{-1} were observed after both SMIT and HIT interventions in the current study
23 respectively.

24 The positive effects of HIT intervention on rowing time trial performance have been
25 demonstrated in two studies (8, 9). 7% improvement of VO_2Peak was reported after 4 weeks

of HIT (8). In row terms, 8 sec (8) and 6 sec (9) improvements have been reported in 2000 m rowing time trial performance following the 4-weeks HIT intervention. In the current study, 2000 m time trial performances were improved 5.7 s and 5 s following SMIT and HIT interventions respectively. Similar to the findings of the current study, the performance improvements that occurred following the 4-week HIT intervention have been found to be associated with the enhancements in the VO_2Peak values, PPO and 4 mmol.L^{-1} power (8, 9).

In the design of the current study, 8 training sessions were completed for each training intervention. There are studies that report improvements by 3.5% in 40 km cycling time trial performance following 6, 8 and 12 HIT sessions (15, 24). It appears that increasing the number of HIT sessions does not result in any further performance improvement in cycling performance. Besides, improvements in 40 km cycling time trial performance related to HIT intervention were reported to be completed after only 6 sessions (14). Also, whether the trend associated with the number of SMIT or HIT sessions into a rowing training program is similar still remains unanswered and may be an interesting area of future research. Furthermore, determining optimal number of the sets and rest intervals during SMIT is another factor to research on.

PRACTICAL APPLICATIONS

In a practical perspective for rowing coaches, about 6 s improvement in 2000 m time trial performance were observed following SMIT intervention, and also 5 s improvement were observed to be associated with HIT intervention. Improvement of about 6 s is equal to approximately a 2 boat length improvement in a single sculling race and these improvements

1 can be decisive factors in the race result. Both methods proved to be successful in improving
2 rowing time trial performance, the benefits gained following SMIT intervention were similar
3 compared with HIT intervention in lightweight collegiate rowers. It should be noted that these
4 outcomes were observed in national level collegiate rowers. The question about the effects of
5 SMIT on performances of elite level rowers remains to be unanswered. Furthermore,
6 considering that the current study is the first to show performance improvements after 4
7 weeks of SMIT intervention in rowers, more research is needed to determine the optimal
8 volume and periodization of the SMIT interventions in a rowing training program.

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1
2 **FIGURE CAPTIONS**
3

4 **Figure 1.** Changes as a percentage in measured variables after supramaximal interval training
5 (SMIT) and high intensity interval training (HIT) interventions.
6
7

ACCEPTED

Table 1. 2000 m. Time Trial and Incremental Exercise Test Performances before and after the Training Interventions and effect of Supramaximal Interval Training (SMIT) relative to High Intensity Interval Training (HIT) and Likelihoods of Clinically Substantial Differences.

Variable	SMIT		HIT		Likelihood (%) of SMIT Being Positive/Trivial/Negative (compared with HIT)
	Pre	Post	Pre	Post	
2000 m time (s)	412 ±7.7	406.3* ± 7.1	411.6 ± 7.5	406.6* ± 7	39% / 31% / 30%
2000 m power (W)	320 ±24.6	335* ± 26.2	322 ±25	334* ± 26	36% / 38% / 26%
Vo_{2peak} (mL.kg⁻¹.min⁻¹)	56.1 ± 5.5	59.2* ± 5.8	57 ± 5.7	59.8* ± 5.4	34% / 40% / 26%
Vo_{2peak} (L.min⁻¹)	4.08 ± 0.66	4.30* ± 0.61	4.10 ± 0.65	4.29* ± 0.63	32% / 39% / 29%
PPO (W)	335 ± 24	353* ± 26	336 ± 20	351* ± 21	44% / 38% / 18%
4 mmol.L⁻¹ power (W)	238 ± 27	256* ± 28	240 ± 23	257* ± 22	45% / 41% / 14%

* p< 0.05 (between pre and post measurements of each group).

